Huawei Access Node QoS Congestion Event Resolution Procedure for KR-ACC-72

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8412

Description

This Standard Operating Procedure (SOP) outlines the steps to identify and resolve QoS congestion events on a Huawei Access Node. These events are typically triggered by microbursts, interface tail drops, and QoS queue starvation. Understanding the underlying protocol mechanics is crucial for resolving these issues. Huawei's QoS framework relies on the Differentiated Services (DiffServ) model, which prioritizes traffic based on the DSCP (Differentiated Services Code Point) values. However, in cases of severe congestion, the interface tail drops packets, leading to QoS queue starvation and subsequent alarms. The Role-Based Access Control (RBAC) feature in Huawei's equipment ensures that only authorized personnel can configure and monitor QoS settings. Hence, the failure to configure QoS settings correctly or the excessive utilization of QoS resources can lead to congestion events. In this scenario, the KR-ACC-72 node is the affected Access node, located at the KR Puram Access site, running on Huawei CLI dialect.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure correct identification of the affected Access node., reference: Refer to the Huawei Access Node documentation for more information.

- check: Check the redundant link status to ensure that it is functioning correctly., reference: Verify the link status using the show interface command.

- check: Dispatch remote hands to the KR Puram Access site for assistance if necessary., reference: Coordinate with the site team to ensure safe access to the equipment.

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops due to excessive packet transmission.	High
QoS queue starvation due to incorrect configuration settings.	Medium
Hardware failure or malfunction of the Access node.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue-info, Expected Output: Queue information: queue 1, packet drops 1000, queue utilization 80%, Description: This command displays the QoS queue information, including packet drops and queue utilization. It helps identify any queue starvation issues.

Command: display interface , Expected Output: Interface , operational status up, input rate 1000 kbps, output rate 500 kbps, Description:
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```
This command displays the interface information, including operational status, input and output rates. It helps identify any interface tail drop issues.

Command: display qos policer-info, Expected Output: Policer information: policer 1, committed rate 1000 kbps, excess rate 500 kbps, Description: This command displays the QoS policer information, including committed and excess rates. It helps identify any QoS configuration issues.
```

Resolution Steps

Step 1: Step: Configure QoS settings to prevent queue starvation, Command: qos queue 1 queue-size 1000, Description: This command sets the queue size to 1000, preventing queue starvation and potential packet drops.

Step 2: Step: Adjust the interface transmission rate to prevent microbursts, Command: interface rate-limit 1000 kbps, Description: This command adjusts the interface transmission rate to 1000 kbps, preventing microbursts and potential interface tail drops.

Step 3: Step: Verify QoS policer configuration, Command: display qos policer-info, Description: This command verifies the QoS policer configuration, ensuring that it is correctly set to prevent excessive packet transmission.

Post-Check Verification

```
Command: display qos queue-info, Expected Output: Queue information: queue 1, packet drops 0, queue utilization 20%, Description: This command verifies the QoS queue information, ensuring that there are no packet drops and the queue utilization is within normal limits.

Command: display interface , Expected Output: Interface , operational status up, input rate 1000 kbps, output rate 500 kbps, Description: This command verifies the interface information, ensuring that it is operational and transmitting at the correct rates.

Command: display qos policer-info, Expected Output: Policer information: policer 1, committed rate 1000 kbps, excess rate 500 kbps, Description: This command verifies the QoS policer information, ensuring that it is correctly configured to prevent excessive packet transmission.
```

Roll-back Procedures

- Step: Restore original QoS queue settings, Command: undo qos queue 1 queue-size, Description: This command restores the original QoS queue settings, reverting to the previous configuration.

- Step: Restore original interface transmission rate, Command: undo interface rate-limit, Description: This command restores the original interface transmission rate, reverting to the previous configuration.

- Step: Verify QoS configuration, Command: display qos policer-info, Description: This command verifies the QoS configuration, ensuring that it is correctly set and configured to prevent excessive packet transmission.